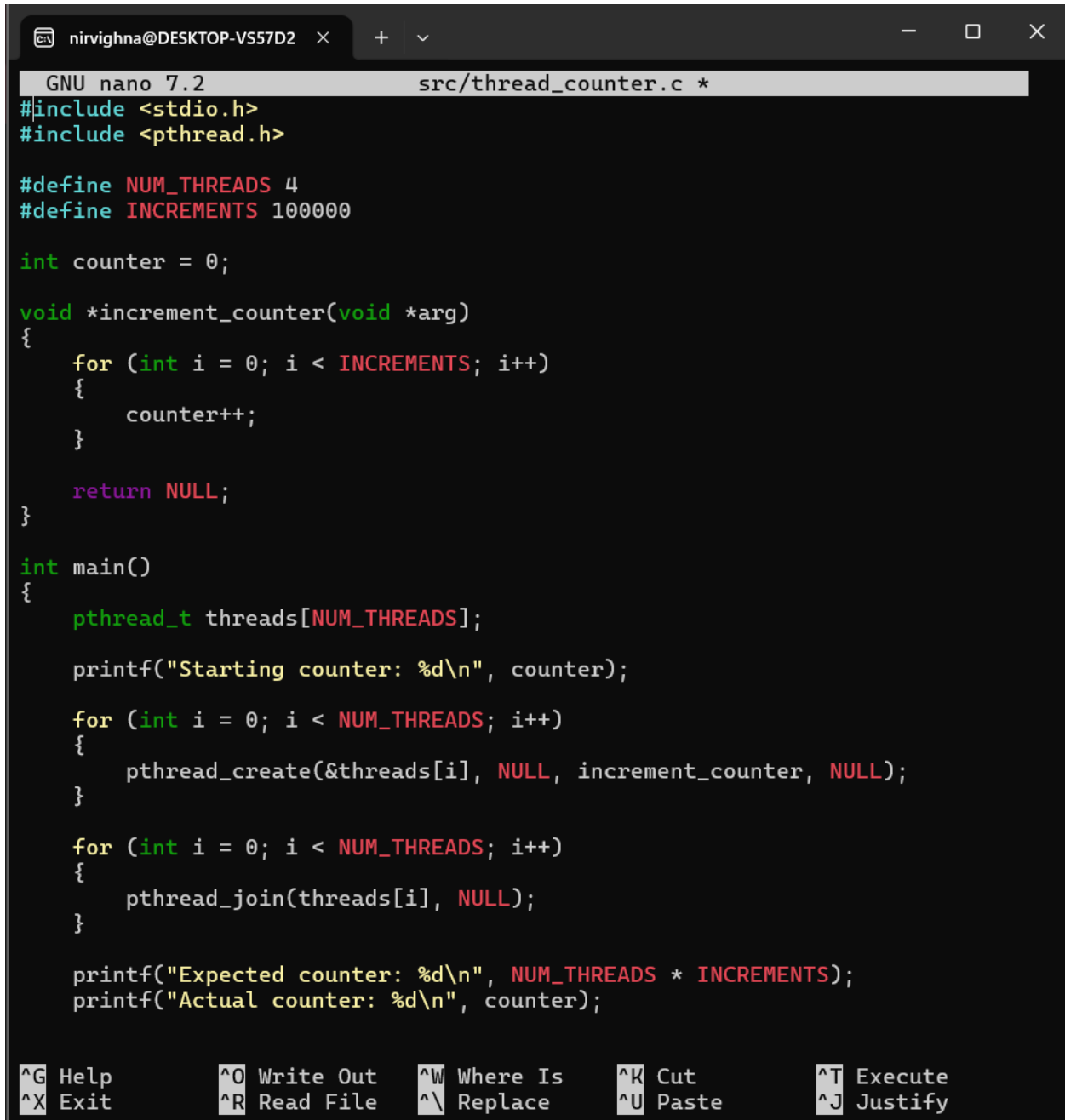


PRACTICAL 11:



```
GNU nano 7.2 src/thread_counter.c *
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define INCREMENTS 100000

int counter = 0;

void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        counter++;
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    printf("Starting counter: %d\n", counter);

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment_counter, NULL);
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("Expected counter: %d\n", NUM_THREADS * INCREMENTS);
    printf("Actual counter: %d\n", counter);
}
```

Help **Write Out** **Where Is** **Cut** **Execute**
Exit **Read File** **Replace** **Paste** **Justify**

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l
total 156
-rw-r--r-- 1 nirvighna nirvighna 498 Sep 21 05:14 Makefile
-rw-r--r-- 1 nirvighna nirvighna 0 Sep 28 04:17 a.c
-rw-r--r-- 1 nirvighna nirvighna 0 Sep 28 04:17 b.c
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 29 19:04 bin
-rw-r--r-- 1 nirvighna nirvighna 0 Sep 28 04:17 c.c
-rw-r--r-- 2 nirvighna nirvighna 36 Sep 29 18:53 hard_link.txt
-rw-r--r-- 2 nirvighna nirvighna 36 Sep 29 18:53 inode_test.txt
-rw-r--r-- 1 nirvighna nirvighna 60 Sep 29 18:32 input.txt
-rw-r--r-- 1 nirvighna nirvighna 1940 Sep 21 04:00 mini_shell.c
-rw-r--r-- 1 nirvighna nirvighna 100 Sep 29 19:02 mmap_test.txt
-rw----- 1 nirvighna nirvighna 1193 Sep 25 13:34 nano.2750.save
-rw-r--r-- 1 nirvighna nirvighna 60 Sep 29 18:33 output.txt
-rwxr-xr-x 1 nirvighna nirvighna 16352 Sep 28 04:13 pipeline1
-rw-r--r-- 1 nirvighna nirvighna 1119 Sep 28 04:13 pipeline1.c
-rwxr-xr-x 1 nirvighna nirvighna 16352 Sep 28 04:17 pipeline2
-rw-r--r-- 1 nirvighna nirvighna 1868 Sep 28 04:17 pipeline2.c
-rwxr-xr-x 1 nirvighna nirvighna 16440 Sep 25 18:02 producer_consumer
-rw-r--r-- 1 nirvighna nirvighna 1133 Sep 25 18:02 producer_consumer.c
-rw-r--r-- 1 nirvighna nirvighna 91 Sep 29 18:46 redirect_output.txt
lrwxrwxrwx 1 nirvighna nirvighna 14 Sep 29 18:54 soft_link.txt -> inode
est.txt
drwxr-xr-x 2 nirvighna nirvighna 4096 Sep 29 19:04 src
-rw-r--r-- 1 nirvighna nirvighna 60 Sep 29 18:42 stdio_output.txt
-rw-r--r-- 1 nirvighna nirvighna 746 Sep 25 17:34 strace_output.txt
-rw-r--r-- 1 nirvighna nirvighna 0 Sep 28 04:14 test1.txt
-rw-r--r-- 1 nirvighna nirvighna 0 Sep 28 04:14 test2.txt
-rw-r--r-- 1 nirvighna nirvighna 0 Sep 28 04:14 test3.txt
-rwxr-xr-x 1 nirvighna nirvighna 16352 Sep 25 17:48 wait_demo
-rw-r--r-- 1 nirvighna nirvighna 1382 Sep 25 17:47 wait_demo.c
-rwxr-xr-x 1 nirvighna nirvighna 16216 Sep 25 17:52 zombie
-rw-r--r-- 1 nirvighna nirvighna 625 Sep 25 17:51 zombie.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ nano src/thread_counter.c
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/thread_counter
Starting counter: 0
Expected counter: 400000
Actual counter: 400000
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

```
nirvighna@DESKTOP-VS57D2  ×  +  v
GNU nano 7.2 src/thread_counter_mutex.c *
#include <stdio.h>
#include <pthread.h>

#define NUM_THREADS 4
#define INCREMENTS 100000

int counter = 0;

pthread_mutex_t counter_mutex = PTHREAD_MUTEX_INITIALIZER;

void *increment_counter(void *arg)
{
    for (int i = 0; i < INCREMENTS; i++)
    {
        pthread_mutex_lock(&counter_mutex);

        counter++;

        pthread_mutex_unlock(&counter_mutex);
    }

    return NULL;
}

int main()
{
    pthread_t threads[NUM_THREADS];

    printf("Starting counter: %d\n", counter);

    for (int i = 0; i < NUM_THREADS; i++)
    {
        pthread_create(&threads[i], NULL, increment_counter, NULL);
    }

    for (int i = 0; i < NUM_THREADS; i++)
    {
        ^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
    }
}
```

```
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ./bin/thread_counter_mutex
Starting counter: 0
Expected counter: 400000
Actual counter: 400000
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l src/thread_counter*
-rw-r--r-- 1 nirvighna nirvighna 680 Sep 29 19:13 src/thread_counter.c
-rw-r--r-- 1 nirvighna nirvighna 876 Sep 30 03:09 src/thread_counter_mutex.c
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ ls -l bin/thread_counter*
-rwxr-xr-x 1 nirvighna nirvighna 18288 Sep 29 19:14 bin/thread_counter
-rwxr-xr-x 1 nirvighna nirvighna 19480 Sep 30 03:09 bin/thread_counter_mutex
nirvighna@DESKTOP-VS57D2I:~/OS-Week1$ |
```

